

IT3122 Cloud Application Development Project (25%)

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Admin No: 224149Q

Module Group: Group 1

Credentials to your AWS account:

Account ID: 312171776467

User: Default User

In this project, you will be developing a lost-and-found website for NYP using AWS Cloud Services.

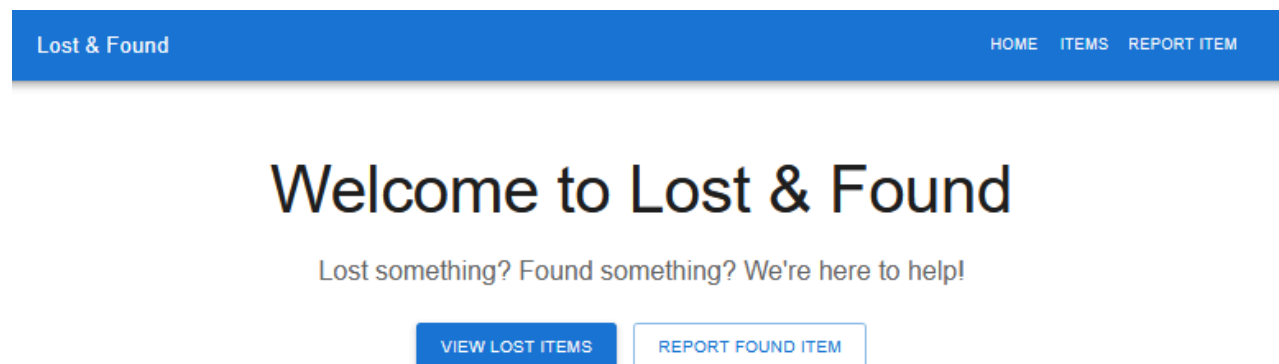
This website will display all items that have been returned to the lost-and-found department.

Part 1 – Setup and host a lost-and-found website (7%)

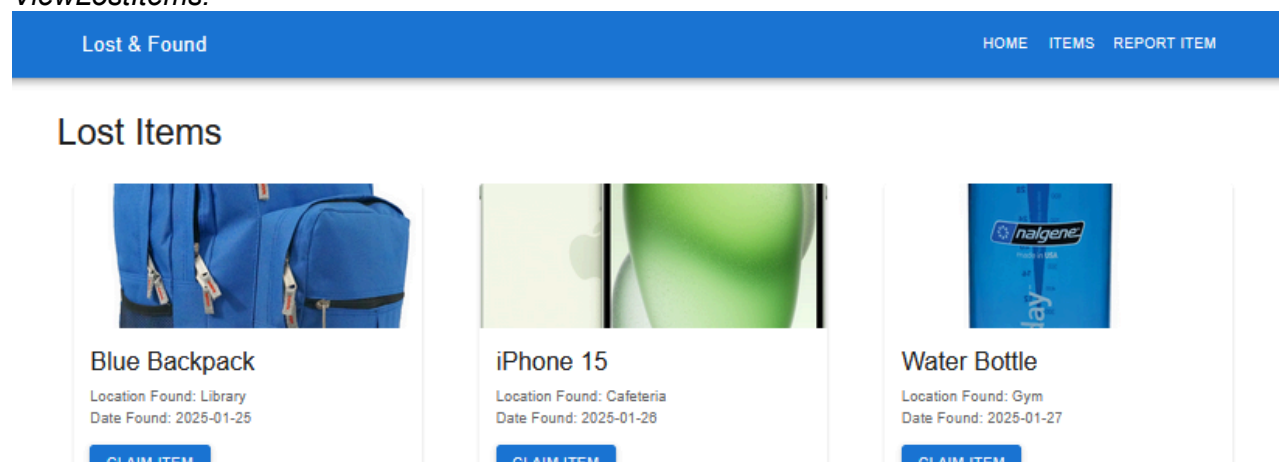
Create a website that allows visitors to the site to display all items that have been returned to the lost-and-found department. You should design your website to be user-friendly and easy to navigate.

(a) Screenshot your website – listing the various pages that you created

HomePage.jsx:



ViewLostItems:



ReportItem:

Lost & Found
HOME ITEMS REPORT ITEM

Report Found Item

Item Name *

Location Found *

Date Found *
dd / mm / yyyy

Description

SUBMIT REPORT

(b) Zip up all the source code for website (label Part 1)

Part 2 – Upload lost items to list on website (5%)

When users find new lost items, they bring them to the lost-and-found counter and these new found items will be added to the website.

As more and more items are found, the lost-and-found department decided to design and streamline the process.

New items can be added to the site, by uploading images of the found items with the appropriate descriptions. A database is now needed to maintain and categorize these items for easy searching and listing.

Design and develop your improved website.

(c) Screenshot your website – listing the various pages that you created and showing all the features of the improved website

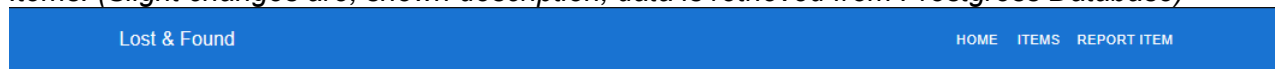
ReportItem: (Now has an Upload Image button which is required before submitting report)

Lost & Found
HOME ITEMS REPORT ITEM

Report Found Item

Item Name *

Items: (Slight changes are, shown description, data is retrieved from PostgreSQL Database)



Lost Items



IPhone1

Location Found: Canteen
 Date Found: 02/02/2025

Description: Found it

[CLAIM ITEM](#)



BAG

Location Found: Library
 Date Found: 01/02/2025

Description: Found it on a chair...alone

[CLAIM ITEM](#)

(d) Screenshot of the database design (you may use SQL or NoSQL)

Using PostgreSQL in AWS RDS:

items	
id 🔗	integer
name 📄	varchar(255) NN
location 📄	varchar(255) NN
date_found 📄	date NN
description 📄	text
image_url 📄	varchar(255)
created_at 📄	timestamp NN

(e) Write a brief explanation on the choice of database used and the design.

I chose PostgreSQL for my database over other alternatives like Aurora MySQL, Aurora PostgreSQL and MySQL as the normal PostgreSQL provides a bit better and easier management. The other types of databases have advanced features like auto-scaling and read replicas which I would not particularly need for this project, as of right now. PostgreSQL's superior ACID compliance and rich data type support also make it ideal for the scope of the project as of right now. Plus, I haven't tried using PostgreSQL before and would like to have a go at it.

The database design is quite normal and standard as should I say:

ID (auto-incrementing primary key)

- Name of the found item
- Location where it was found
- Date when found
- Description
- Image URL (storing S3 links)
- Creation timestamp

This database structure supports the core functionality of recording and displaying items, with all data properly organized and easily accessible.

(f) Zip up all the source code for website and database (label Part 2)

Part 3 (8%)

As the site grew, more and more items were found and claimed and traffic to the site also increased by a lot. The team decided to make it easier for users to find their lost items.

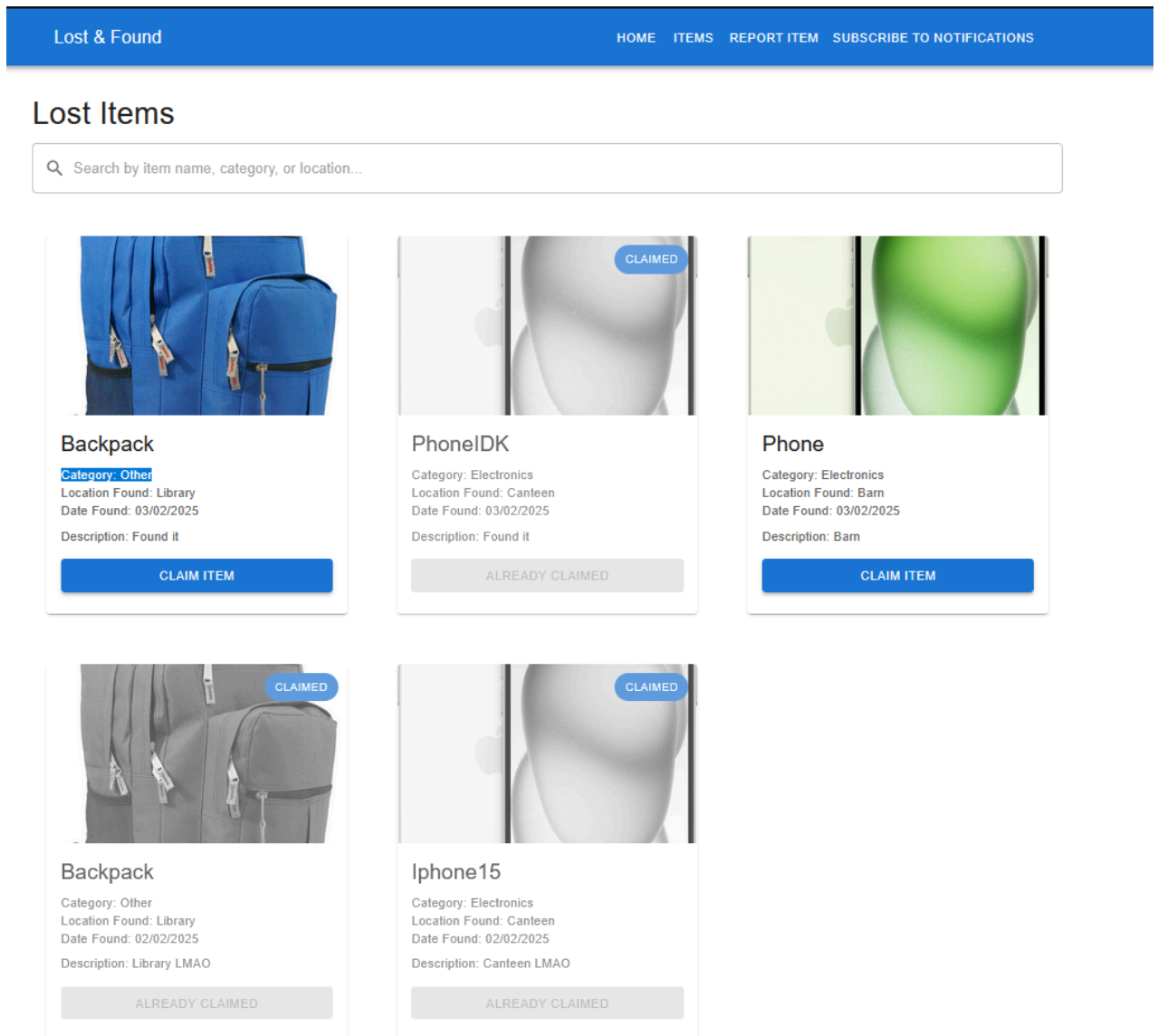
Users can create notification whenever new items are listed on the site or when an item that matched the category of item is uploaded to the site.

The team was also exploring other features to improve user experience and automation (eg using AI to recognize items etc). The team also decided to improve the site by improving availability and reliability.

Design and utilize any AWS service to achieve this feature.

- (g) Screenshot the coding of your improvements and the respective settings. Show evidence that these features are working.

New Items outlook:



Lost & Found

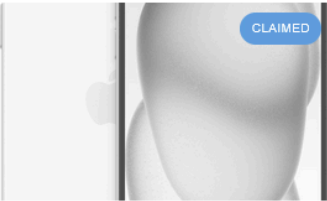
HOME ITEMS REPORT ITEM SUBSCRIBE TO NOTIFICATIONS

Lost Items

Search by item name, category, or location...



Backpack
 Category: Other
 Location Found: Library
 Date Found: 03/02/2025
 Description: Found it
 CLAIM ITEM



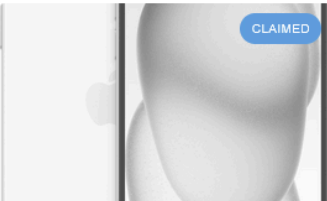
PhoneIDK
 Category: Electronics
 Location Found: Canteen
 Date Found: 03/02/2025
 Description: Found it
 ALREADY CLAIMED



Phone
 Category: Electronics
 Location Found: Barn
 Date Found: 03/02/2025
 Description: Barn
 CLAIM ITEM



Backpack
 Category: Other
 Location Found: Library
 Date Found: 02/02/2025
 Description: Library LMAO
 ALREADY CLAIMED



Iphone15
 Category: Electronics
 Location Found: Canteen
 Date Found: 02/02/2025
 Description: Canteen LMAO
 ALREADY CLAIMED

New Upload page:

Lost & Found

HOMEITEMSREPORT ITEMSUBSCRIBE TO NOTIFICATIONS

Report Found Item

1 Upload Image

2 Review AI Analysis

3 Submit Report

UPLOAD IMAGE

With AI recognition:

Lost & Found

HOMEITEMSREPORT ITEMSUBSCRIBE TO NOTIFICATIONS

Report Found Item

✓ Upload Image

2 Review AI Analysis

3 Submit Report

Image Preview



AI Analysis Results

Detected Objects:

Electronics (98.85%)

Phone (94.76%)

Mobile Phone (93.89%)

Speaker (83.45%)

Egg (80.11%)

Food (80.11%)

Suggested Category: Electronics

CONTINUE TO REPORT

Lost & Found

HOMEITEMSREPORT ITEMSUBSCRIBE TO NOTIFICATIONS

Report Found Item

✔ Item successfully reported! Notifications will be sent to subscribed users.

✔ Upload Image

✔ Review AI Analysis

✔ Submit Report

Confirm Claim:

Lost & Found

HOMEITEMSREPORT ITEMSUBSCRIBE TO NOTIFICATIONS

Lost Items

Search by item name, category, or location...



PhoneAndroid

Category: Electronics
Location Found: Library
Date Found: 03/02/2025
Description: Found it

CLAIM ITEM

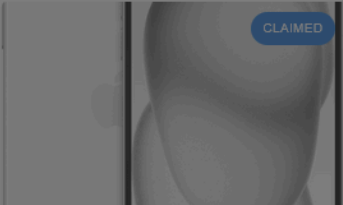


Backpack

Confirm Claim

Are you sure you want to claim this item?
Item: PhoneAndroid
Category: Electronics
Location: Library

CANCELCONFIRM CLAIM



PhoneIDK

Category: Electronics
Location Found: Canteen
Date Found: 03/02/2025
Description: Found it

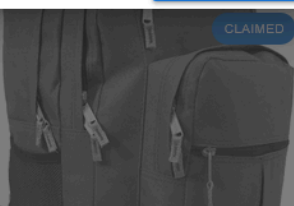
ALREADY CLAIMED



Phone

Category: Electronics
Location Found: Barn
Date Found: 03/02/2025
Description: Barn

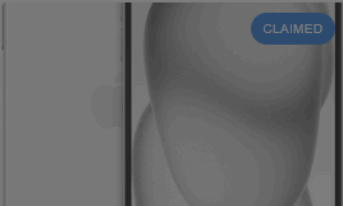
CLAIM ITEM



Backpack

Category: Other
Location Found: Library
Date Found: 02/02/2025
Description: Library LMAO

ALREADY CLAIMED



Iphone15

Category: Electronics
Location Found: Canteen
Date Found: 02/02/2025
Description: Canteen LMAO

ALREADY CLAIMED

Subscribe to notifications:

Notification Subscriptions

Subscribe to receive email notifications when new items are found in your categories of interest.

Email *

You will receive a confirmation email to verify your subscription

Category *

Select a category to receive notifications for

SUBSCRIBE

Notification Subscriptions

Subscribe to receive email notifications when new items are found in your categories of interest.

✔

Subscription pending. Please confirm the subscription in your email.

Email *

khai-aws@8mpty.com

You will receive a confirmation email to verify your subscription

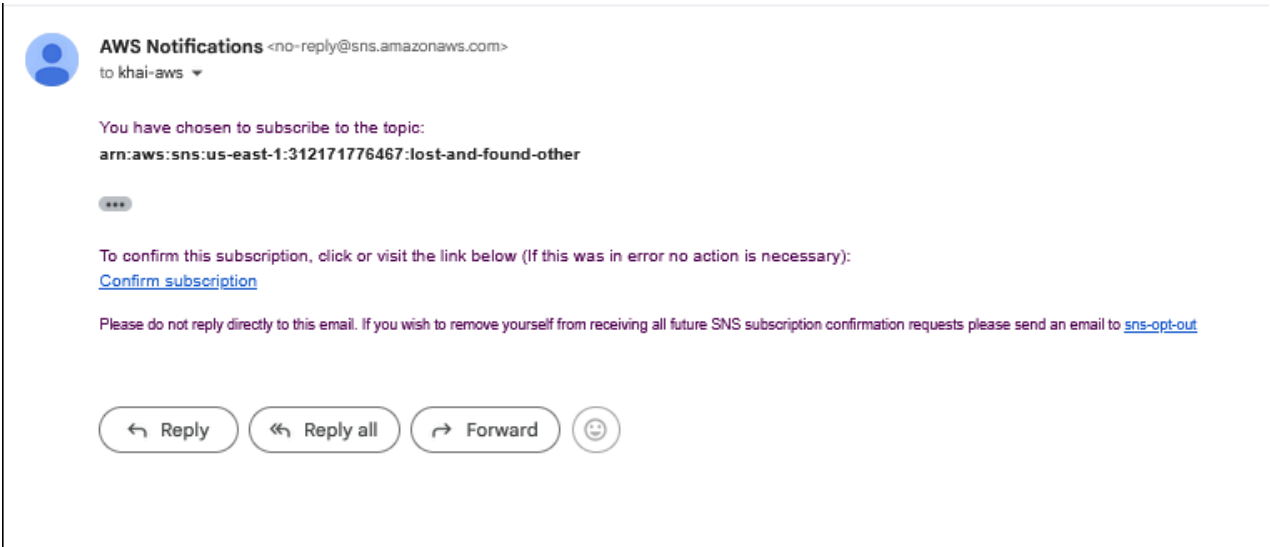
Category *

Select a category to receive notifications for

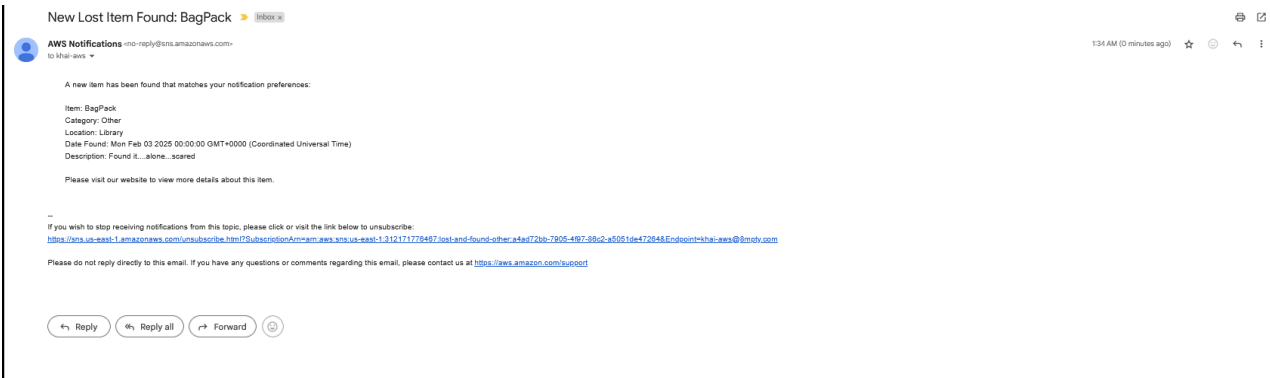
SUBSCRIBE

Your Active Subscriptions

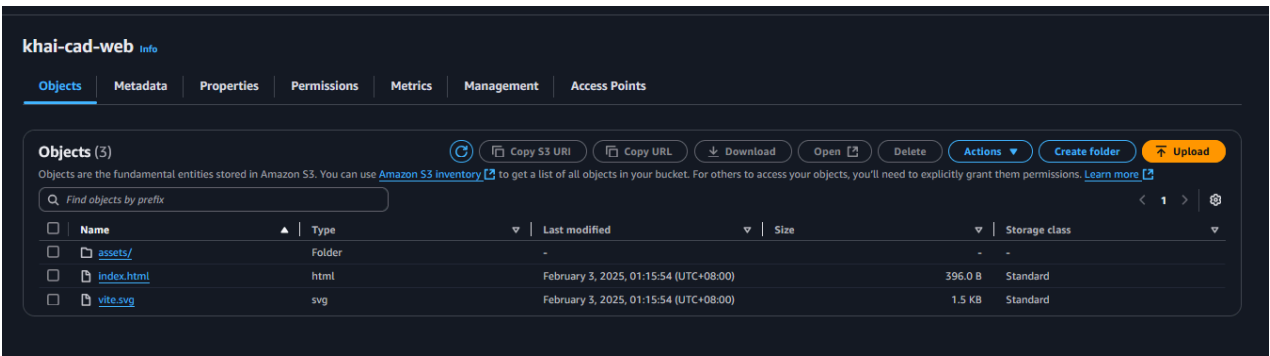
Other



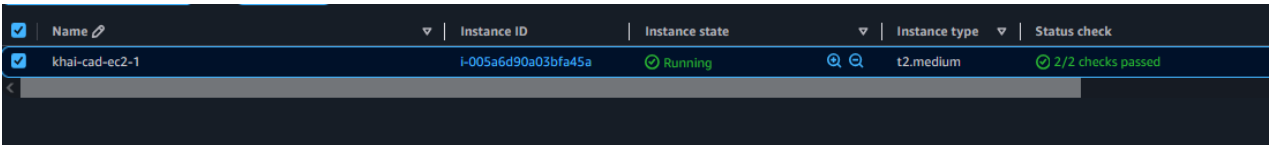
Email Notification:



Used S3 to host frontend:



EC2 for Backend:

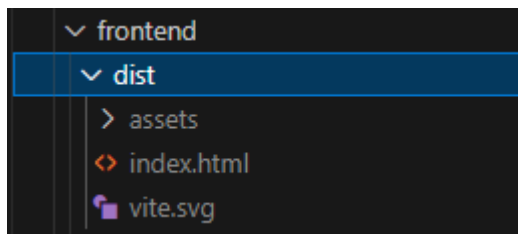
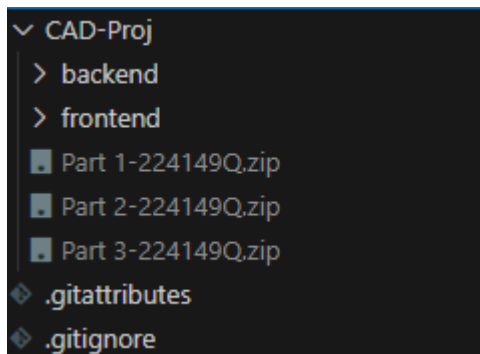


- (h) Document your design and steps with the necessary descriptions and screenshots.

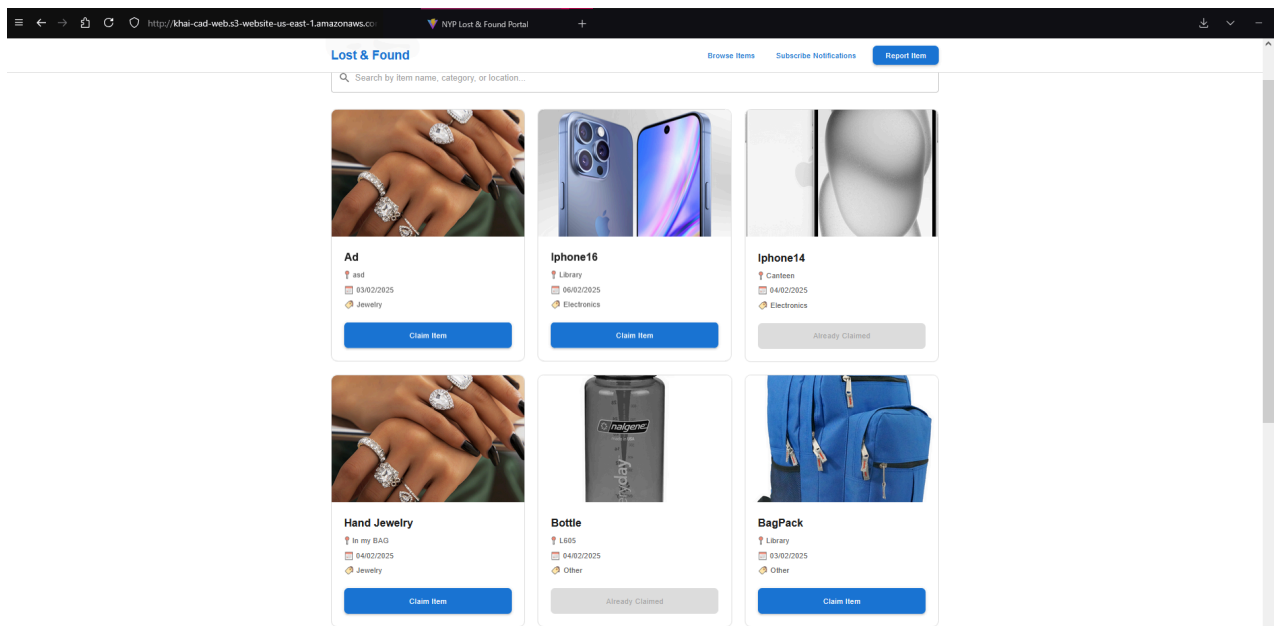
So for my project, I have a backend and frontend. The backend is based on NodeJs while the frontend is based on ReactJS. I decided to then host my application onto the cloud so that way more users around the world could access my web application.

My codes are on Github so that I could clone my project into an EC2 instance that would hold and deploy the application. I used the EC2 only for the backend while the frontend is just the "build" files from ReactJS using "npm run build" and uploaded to a S3 with public access enabled.

The backend, like I said, is hosted on EC2. It is also built and deployed using docker so that it automatically starts the container and backend when the instance ever restarts. I have also used an AI functionality called Amazon Rekognition which would analyze the Uploaded image and "predicts/recommends" what category the item would belong to. Users are now also able to subscribe to a SNS topic which would send them an email notification whenever the desired category has a new item.



Minor updates to the UI and hosted to S3 and EC2:



```

ubuntu@ip-172-31-19-177: ~
PS C:\Users\EmptyDesktop\Downloads> ssh -i .\labsuser.pem ubuntu@52.5.74.1
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Feb  8 17:52:50 UTC 2025

System load:  0.0          Processes:           115
Usage of /:   45.0% of 6.71GB Users logged in:       0
Memory usage: 32%          IPv4 address for enx0: 172.31.19.177
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

57 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Feb  8 17:52:43 2025 from 182.19.251.80
ubuntu@ip-172-31-19-177:~$ ls -ll
total 4
drwxr-xr-x 4 root root 4096 Feb  3 19:14 CAD-Proj
ubuntu@ip-172-31-19-177:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED    STATUS    PORTS                               NAMES
3a1a157032b2   cad-backend    "docker-entrypoint.s..." 4 days ago Up About an hour 0.0.0.0:3001->3001/tcp, :::3001->3001/tcp cad-backend
ubuntu@ip-172-31-19-177:~$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED    SIZE
cad-backend   latest    2d06d015db45   4 days ago 146MB
ubuntu@ip-172-31-19-177:~$
  
```

Part 4 (5%) – Advanced automation and features

The lost-and-found project was a great success. Other polytechnics heard about this project and have approached your team for assistance in implementing a similar project.

Design, document and implement your automation plan.

Your team had other plans and ideas for enhancement and improvement to the site, these enhancements should be included

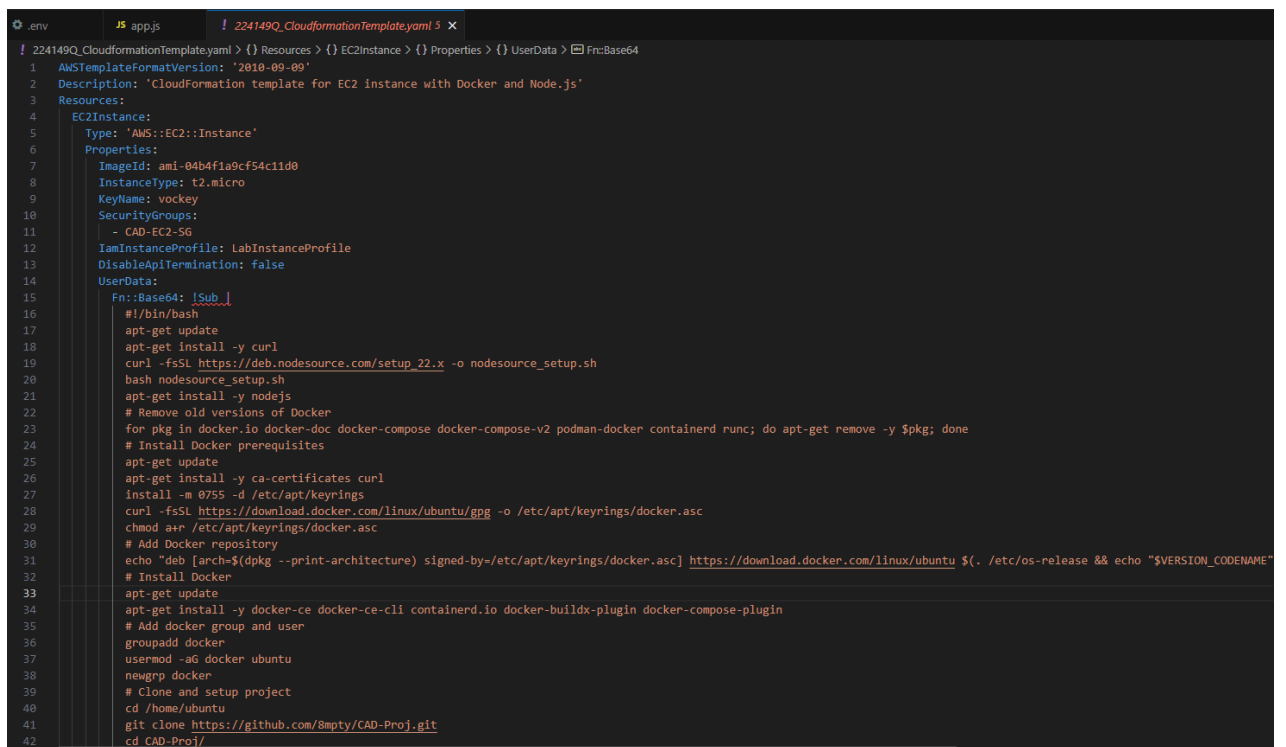
- (i) Document your design and steps with the necessary descriptions and screenshots for the automation process.

I implemented the cloudformation stack where it would create the elastic IP (for easier IP assignment), deploy an EC2 instance which would install all the dependencies needed like nodeJS and docker and finally git clone my project. I too have upload the Cloudformation template to the github repo.

After the stack has successfully been created, i would then ssh into the server to build the container. This could normally be done automatically with the docker file but there are environment variables which would need to be changed/updated to match the scope of the project. The docker file is also uploaded to the github repo. There were minor updates to the UI so that it matches the look of a more modern looking website.

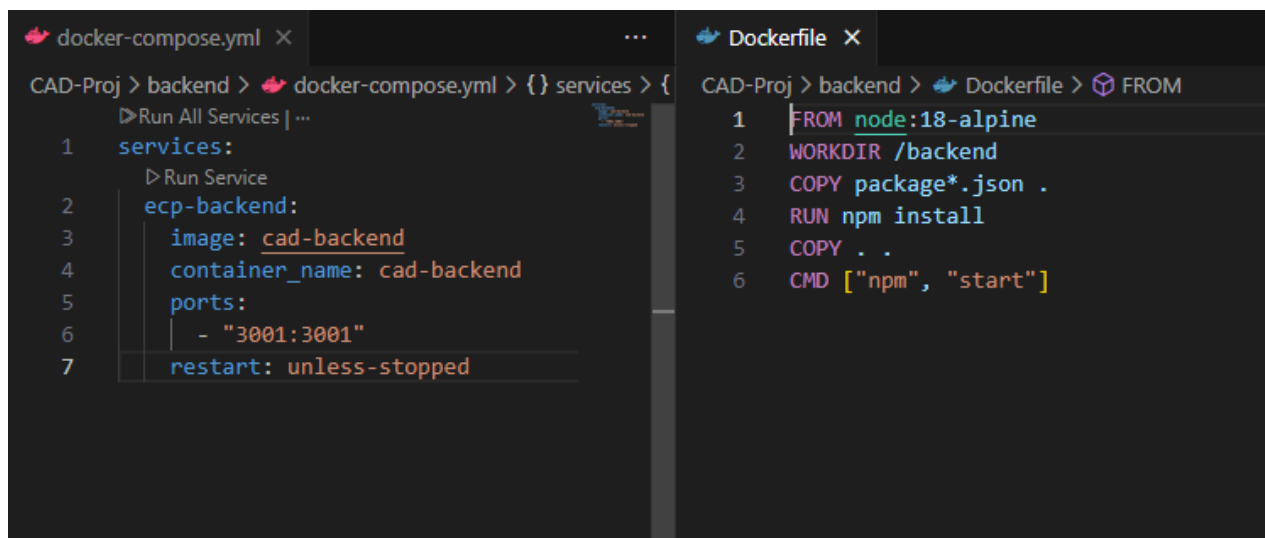
The project can be viewed here: (<https://github.com/8mpty/CAD-Proj/tree/dev>)

Website: (<http://khai-cad-web.s3-website-us-east-1.amazonaws.com>)



```

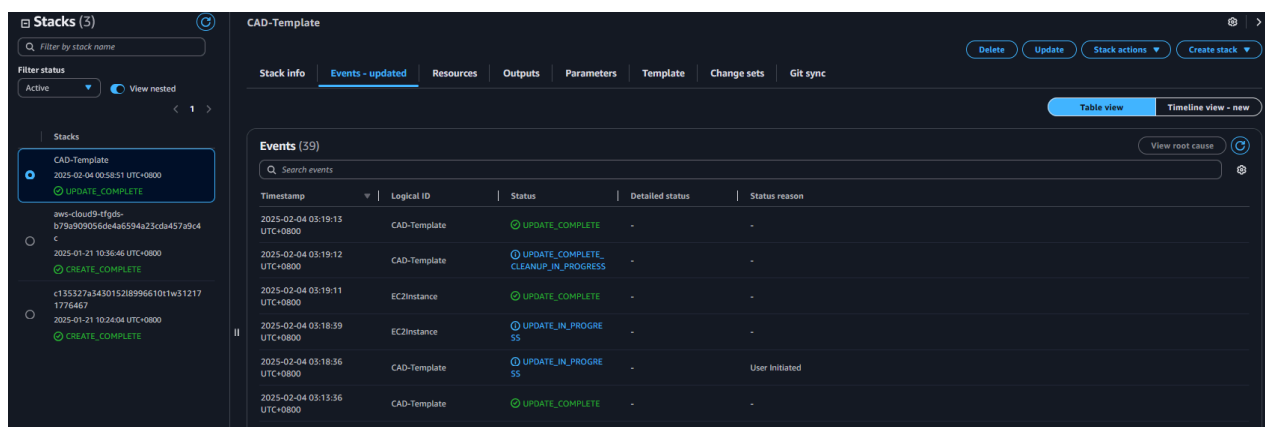
! 224149Q_CloudFormationTemplate.yaml 5 x
1 AWSTemplateFormatVersion: '2010-09-09'
2 Description: 'CloudFormation template for EC2 instance with Docker and Node.js'
3 Resources:
4   EC2Instance:
5     Type: 'AWS::EC2::Instance'
6     Properties:
7       ImageId: ami-04b4f1a9cf54c11d0
8       InstanceType: t2.micro
9       KeyName: vockey
10      SecurityGroups:
11        - CAD-EC2-SG
12      IamInstanceProfile: LabInstanceProfile
13      DisableApiTermination: false
14      UserData:
15        Fn::Base64: !Sub [
16          #!/bin/bash
17          apt-get update
18          apt-get install -y curl
19          curl -fsSL https://deb.nodesource.com/setup_22.x -o nodesource_setup.sh
20          bash nodesource_setup.sh
21          apt-get install -y nodejs
22          # Remove old versions of Docker
23          for pkg in docker.io docker-doc docker-compose docker-compose-v2 podman-docker containerd runc; do apt-get remove -y $pkg; done
24          # Install Docker prerequisites
25          apt-get update
26          apt-get install -y ca-certificates curl
27          install -m 0755 -d /etc/apt/keyrings
28          curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
29          chmod a+r /etc/apt/keyrings/docker.asc
30          # Add Docker repository
31          echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "$VERSION_CODENAME"
32          # Install Docker
33          apt-get update
34          apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
35          # Add docker group and user
36          groupadd docker
37          usermod -aG docker ubuntu
38          newgrp docker
39          # Clone and setup project
40          cd /home/ubuntu
41          git clone https://github.com/8mpty/CAD-Proj.git
42          cd CAD-Proj/
  
```



```

docker-compose.yml
CAD-Proj > backend > docker-compose.yml > {} services > {
  1 services:
  2   ecp-backend:
  3     image: cad-backend
  4     container_name: cad-backend
  5     ports:
  6       - "3001:3001"
  7     restart: unless-stopped

Dockerfile
CAD-Proj > backend > Dockerfile > FROM
  1 FROM node:18-alpine
  2 WORKDIR /backend
  3 COPY package*.json .
  4 RUN npm install
  5 COPY . .
  6 CMD ["npm", "start"]
  
```



Stacks (5)

Filter by stack name

Filter status: Active View nested

Stacks

- CAD-Template
 - UPDATE_COMPLETE
- aws-cloud9-4fgds-079a909056de4a6594a23cda457a9c4
 - CREATE_COMPLETE
- c135337a34301521899661011w312171776447
 - CREATE_COMPLETE

CAD-Template

Stack info Events - updated Resources Outputs Parameters Template Change sets Git sync

Table view Timeline view - new

Events (39)

Search events

Timestamp	Logical ID	Status	Detailed status	Status reason
2025-02-04 03:19:13 UTC+0800	CAD-Template	UPDATE_COMPLETE	-	-
2025-02-04 03:19:12 UTC+0800	CAD-Template	UPDATE_COMPLETE_CLEANUP_IN_PROGRESS	-	-
2025-02-04 03:19:11 UTC+0800	EC2Instance	UPDATE_COMPLETE	-	-
2025-02-04 03:18:39 UTC+0800	EC2Instance	UPDATE_IN_PROGRESS	-	-
2025-02-04 03:18:36 UTC+0800	CAD-Template	UPDATE_IN_PROGRESS	-	User Initiated
2025-02-04 03:13:36 UTC+0800	CAD-Template	UPDATE_COMPLETE	-	-

(j) Write a brief explanation on the future enhancement plan of your lost-and-found project.

For the future enhancement, I could definitely include the use of a Login form so that there could be management on the Admin side and the User side. The basic could then have it so that only the Admins are able to upload/delete/update the items while the users can only view or claim such items.

Appendix – Rubrics

CATEGORY	A	B	C	D
Implementation (45%)	Completed all system with persistence storage integration and all managed cloud service that is aligned to the problem statement. The system is user friendly and caters to all groups of users fully.	Partial completion of system with persistence storage integration and some managed cloud service that is aligned to the problem statement. The system has some user-friendly features implemented.	Incomplete system with persistence storage integration and minimum or no managed cloud service that is aligned to the problem statement. The system has basic user navigation components.	Incomplete system without persistence storage integration and no managed cloud service
Cloud Native Development (20%)	Demonstrate effective use of at least 5 managed cloud services. All the managed cloud services are fully integrated	Demonstrate effective use of 2 managed cloud service. The cloud services used are partially integrated	Able to identify and use basic cloud service such as compute, storage or database to implement the solution. Some form of integration has been used.	Unable to use any cloud service to implement the system
Infrastructure as Code (15%)	Demonstrate effective use of infrastructure as code to create the cloud infrastructure automatically	Demonstrate partial use of infrastructure as code to create the cloud infrastructure partially	Able to create cloud infrastructure using the management console	Unable to completely create the cloud infrastructure
Explanation (20%)	Confident with good illustration and explanation to explain the work well. Logical and feasible proposals	Somewhat confident with decent illustration and explanation to somewhat explain the work done	Lack of confident with some illustration and explanation, but unable to explain the work well	No confidence, and little illustration and explanation, unable to describe the work done

<~The END~>